EXPERIMENT-03

AIM-To get the input matrix from user and perform Matrix addition, subtraction, multiplication, inverse transpose and division operations using vector concept in python.

CODE-

```
import numpy as np

r = int(input("Enter the number of rows: "))
c = int(input("Enter the number of columns: "))

print("Enter the elements of Matrix A:")
A = np.array([[float(input()) for j in range(c)] for i in range(r)])

print("Enter the elements of Matrix B:")
B = np.array([[float(input()) for j in range(c)] for i in range(r)])

print("Matrix A:\n", A)
print("Matrix B:\n", B)

print("Addition (A+B):\n", A + B)
print("Subtraction (A-B):\n", A - B)

# Safe Division
```

```
try:
    print("Division (A/B):\n", A / B)
except:
    print("Division not possible (check for zero values in B)")

# Multiplication
print("Multiplication (A*B):")
if r == c:
    print(np.dot(A, B))
else:
    print("Matrix multiplication not possible")

print("\nTranspose of Matrix A:\n", A.T)

# Inverse
print("\nInverse of Matrix A:")
if r == c:
    try:
        print(np.linalg.inv(A))
    except:
        print("Matrix inverse not possible (Singular Matrix)")
else:
    print("Matrix inverse not possible (Not square)")
```

OUTPUT-

```
Enter the number of rows: 2
Enter the number of columns: 2
Enter the elements of Matrix A:
3
4
5
6
Enter the elements of Matrix B:
7
8
9
1
Matrix A:
 [[3. 4.]
 [5. 6.]]
Matrix B:
 [[7. 8.]
 [9. 1.]]
Addition (A+B):
 [[10. 12.]
 [14.  7.]]
Subtraction (A-B):
 [[-4. -4.]
 [-4.  5.]]
Division (A/B):
 [[0.42857143 0.5       ]
 [0.55555556 6.        ]]
Multiplication (A*B):
[[57. 28.]
 [89. 46.]]

Transpose of Matrix A:
 [[3. 5.]
 [4. 6.]]

Inverse of Matrix A:
[[-3.   2. ]
 [ 2.5 -1.5]]
```